香港資優教育學苑
The Hong Kong Academy for Gifted Education

IMO Training 2023 – 24 Phase I

Theory of Equations (方程理論)

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Quadratic Equation (二次方程)

Consider the equation $ax^2 + bx + c = 0$, where $a \neq 0$

The discriminant (判別式) $\Delta = (b^2 - 4ac)$

(i) If $\Delta = 0$, the two roots are identical

(ii) If Δ is a perfect square, then the roots are rational.

(iii) If $\Delta > 0$, the two roots are real and unequal, and they are equal to

$$\frac{-b-\sqrt{\Delta}}{2a} \text{ a} \quad \text{and} \quad \frac{-b+\sqrt{\Delta}}{2a}$$

Example 1: Show that if a and b are odd integers, the equation $x^2 + 2ax + 2b = 0$ cannot have rational roots.

Example 2: Let a and b be real numbers for which the equation

$$x^4 + ax^3 + bx^2 + ax + 1 = 0$$

has at least one real solution. For all such pairs (a, b), find the minimum value of $a^2 + b^2$. (15th IMO 1973, No. 3)

Relationship between coefficients and roots of a quadratic equation

Consider the equation $ax^2 + bx + c = 0$, where $a \neq 0$

(i) Sum of roots = $-\frac{b}{a}$

(ii) Product of roots = $\frac{c}{a}$.

Example 3 If α and β are the roots of the equation $x^2 - x - 1 = 0$, find the quadratic equation whose roots are $\alpha + \frac{1}{\beta}$ and $\beta + \frac{1}{\alpha}$.

Example 4: Prove that if a, b are integers and b is odd, then the equation $x^2 + 16ax + 7^b = 0$ has no integral solutions

Example 5: For any natural number n, the roots of the equation $x^2 + (2n+1)x + n^2 = 0$ are α_n and β_n. Find the value of

$$\frac{1}{(\alpha_3+1)(\beta_3+1)} + \frac{1}{(\alpha_4+1)(\beta_4+1)} + \dots + \frac{1}{(\alpha_{20}+1)(\beta_{20}+1)} .$$

Example 6: Find all possible values of x, such that for any value of y, the equation $x^2 + y^2 + z^2 + 2xyz = 1$ has at least a solution.

Polynomial Equations (多項方程)

Let $f(x) = a_n x^n + a_{n-1}x^{n-1} + \ldots + a_1 x + a_o$ be a real polynomial in x of degree n.

Next, $f(x) = 0$ is called a real polynomial equation of degree n in x.

Note the following properties

(i) According to the Fundamental Theorem of Algebra and related Theorems,, the equation has n roots, which may be real or complex, and some of them may be double roots, triple roots or roots of multiplicity m..

(ii) If $a_o = 0$, then 0 is one of the roots of the equation.

(iii) If the sum of the coefficients is 0, then 1 is a root of the equation.

(iv) If all the coefficients are positive, then the equation has no positive roots.

(v) If the coefficients are alternatively positive and negative, then the equation has no negative roots.

(vi) If $a_n = 1$, and all the other coefficients are integers, then there cannot be a non-integral rational root.

Example 7: The coefficients of $f(x) = a_n x^n + a_{n-1}x^{n-1} + \ldots + a_1 x + a_o$ are all Integers. Given that p is odd and q is even, and that both f(p) and f(q) are odd. Show that the equation $f(x) = 0$ does not have integral roots.

Example 8: If a and b are rational numbers, can the two equations $x^5 - x - 1 = 0$ and $x^2 + ax + b = 0$ have a common root?

Example 9: If the two quadratic polynomials $P(x) = x^2 + ax + b$ and $Q(x) = x^2 + cx + d$ are such that the equation $P(Q(x)) = Q(P(x))$ does not have a real solution, prove that $b \neq d$.

Example 10: Given $P(x) = 4x^3 - 2x^2 - 15x + 9$, $Q(x) = 12x^3 + 6x^2 - 7x + 1$, prove that

(a) Both $P(x) = 0$ and $Q(x) = 0$ have three distinct real roots.

(b) If α, β are respectively the largest root of $P(x)=0$ and $Q(x) = 0$, then $\alpha^2+3\beta^2 = 4$.

Theorem on Irrational Roots

Theorem: Given that the coefficients of $f(x) = a_n x^n + a_{n-1}x^{n-1} + \ldots + a_1 x + a_o$ are all rational numbers. If $a + \sqrt{b}$ is a root of $f(x) = 0$, then so is $a - \sqrt{b}$.

Proof: Consider $g(x) = (x - a - \sqrt{b})(x - a + \sqrt{b}) = x^2 - 2ax + a^2 - b$

Divide f(x) by g(x) gives $f(x) = (x^2 - 2ax + a^2 - b)\,Q(x) + cx + d$

Put $x = a + \sqrt{b}$, $f(a + \sqrt{b}) = 0$,

therefore $c(a + \sqrt{b}) + d = 0 \Leftrightarrow (ca + d) + c\sqrt{b} = 0$

so $c = 0$, and then $d = 0$

Hence $f(x) = (x^2 - 2ax + a^2 - b)\,Q(x) = (x - a - \sqrt{b})(x - a + \sqrt{b})\,Q(x)$

This means that $a - \sqrt{b}$ is also a root of $f(x) = 0$.

Complex Roots Occur in Conjugate Pairs (虛根成對原理)

Theorem: Let f(x) be a polynomial with real coefficients. If $a + bi$ $(b \neq 0)$ is a root of the equation $f(x) = 0$, then so is $a - bi$.

Proof: $a + bi$ is a root of $f(x) = 0$ => $(x - a - bi)$ is a factor of f(x).

Now divide f(x) by the quadratic expression $(x - a - bi)(x - a + bi) = (x-a)^2 + b^2$

To get $f(x) = (x - a - bi)(x - a + bi)Q(x) + cx + d$

Since $f(a + bi) = 0$, therefore $c(a+bi) + d = 0$, so $ca + d) + cbi = 0$

Equating real and imaginary parts, since $b \neq 0$, therefore $c = 0$, next, $d = 0$.

Hence $f(x) = (x - a - bi)(x - a + bi)Q(x)$, and $a - bi$ is also a root of $f(x) = 0$.

Example 11: Let $x^3 + px^2 + qx + r = 0$ be an equation with real coefficients. If $a + bi$ is a root of $f(x) = 0$, show that $r(a^2 + b^2) = (p + 2a)[(a^2 + b^2)q + 2ar]$

Corollary 1: Every real polynomial $f(x) = a_nx^n + a_{n-1}x^{n-1} + \ldots + a_1x + a_o$ can be broken into monomial and binomial factor of the form

$$f(x) = a_n(x-x_1)\ldots(x-x_m)(x^2+2b_1x+c_1)\ldots(x^2+2b_{\frac{n-m}{2}} + c_{\frac{n-m}{2}})$$

Example 12: Reduce $x^3 - 1$, $x^3 + 1$, $x^4 - 1$, $x^4 + 1$, $x^6 + 1$, $x^6 - 1$, $x^5 + 1$, $x^5 - 1$, $x^7 - 1$ into monomial and quadratic factors

Corollary 2: If a polynomial $f(x)$ is such that the equation $f(x) = 0$ contains only imaginary roots, then $f(x)$ assumes positive values for all values of x.
(Reason: The factors of $f(x)$ are all of the form $(x - a)^2 + b^2$, and they all assume positive values for any value of x.

Corollary 3: If $f(x) = a_nx^n + a_{n-1}x^{n-1} + \ldots + a_1x + a_o$ is a polynomial with rational coefficients and n is odd, then the equation has at least one real root. Further, if the equation has only one real root, then the root must be rational. (Reason: Complex roots occur in pairs. Further, irrational roots also occur in pairs.)

Descartes Rule of Signs (笛卡兒符號法則)

Theorem: Let $f(x)$ be a polynomial of degree n, with rational coefficients. Consider the equation $f(x) = 0$. The number of positive roots of the equation is equal to the number of sign changes, or less than the number of sign changes by a positive even number. (Note: here a root of multiplicity m is counted as m roots.)

Proof: Suppose the equation $f(x) = 0$ possesses p positive roots, namely $\alpha_1, \alpha_2, \alpha_3, \ldots\alpha_p$, q zero roots, and r negative roots namely $-\beta_1, -\beta_2, -\beta_3, \ldots -\beta_r$, together with $2s$ complex roots, $a_1 \pm b_1i$, $a_2 \pm b_2i$, $a_3 \pm b_3i, \ldots.\ a_s \pm b_si$,

Then $f(x) = (x - \alpha_1)(x - \alpha_2)(x - \alpha_3)\ldots\ldots(x - \alpha_p)\,\varphi(x)$, where

$$\varphi(x) = x^q(x + \beta_1)(x + \beta_2)(x + \beta_3)\ldots\ldots(x + \beta_r)(x^2 - 2a_1x + (a_1^2 + b_1^2)).$$
$$(x^2 - 2a_2x + (a_2^2 + b_2^2))\ldots\ldots(x^2 - 2a_sx + (a_s^2 + b_s^2))$$

Note that, when expanded, the first term and last term of $\varphi(x)$ are both positive, hence the number of changes in sign is an even number, say $2k$, where $k = 0, 1, 2, \ldots$ Multiplying $\varphi(x)$ by $(x - \alpha_1)$, $(x - \alpha_2)$, $(x - \alpha_3)$, ……and $(x - \alpha_p)$, each time the number of changes of sign is increased by $2k_i + 1$, where $i = 1, 2, 3, \ldots p$.

We may show the above statement by considering the following:

$\varphi(x)$	+	-	-	+	+	-	-	-	+	-	+	-	+	
$(x - \alpha_i)$	+	-												
$x\varphi(x)$	+	-	-	+	+	-	-	-	+	-	+	-	+	
$(-\alpha_i)\varphi(x)$		-	+	+	-	-	+	+	+	-	+	-	+	-
$(x - \alpha_i)\ \varphi(x)$	+	-	±	+	±	-	±	±	+	-	+	-	+	-

We see that the number of changes in sign is increased by $2k_i + 1$.

Hence the total number of changes in sign is

$2k + (2k_1+1) + (2k_2+1) + (2k_3+1) + \ldots + (2k_p+1) = 2k' + p$

This is the number of changes in the sign of the coefficients in $f(x)$.

The number of positive roots in $f(x) = 0$, is equal to the number of changes in sign (if $k' = 0$), or equal to the number of changes in sign less an even positive number (if $k' \neq 0$).

Corollary 1.: The equation $f(x) = 0$ cannot have more positive roots than there are changes of sign in $f(x)$.

Corollary 2: The equation $f(x) = 0$ cannot have more negative roots than there are changes of sign in $f(-x)$.

Corollary 3: If there are p changes of sign in f(x), and q changes of sign in f(-x), then there are at least $n - p - q$ complex roots.

Example 13: Show that the equation $x^4 + 1 = 0$ has no real roots..

Example 14: Consider $f(x) = x^5 + x^2 + 1$. Show that the equation $f(x) = 0$ has 4 complex roots.

Example 15: Consider $f(x) = x^9 + 5x^8 - x^3 + 7x + 1$. Show that the equation $f(x) = 0$ has at least 4 complex roots.

Example 16: Show that the equation $x^3 - x^2 + 2x + 1 = 0$ does not have any positive root.

Example 17: Show that the equation $2x^3 + 3x^2 - 4x + 5 = 0$ does not have a root which is greater that 1.

Example 18: Find the least number of complex roots of the equation
$x^{3n} - x^{2n} + x^n + x + 1 = 0$.

Example 19**:** Prove that the equation
$x^{2n} - 2x^{2n-1} + 3x^{2n-2} - 4x^{2n-3} + \ldots - 2nx + 2n + 1 = 0$
does not have any real root.

Relationship between coefficients and roots of a polynomial equation
(Vieta's formulae 韋達公式))

Consider the equation $f(x) = 0$

(iii) Sum of roots = $- a_{n-1}/a_n$
(iv) Sum of products of roots taking two at a time = a_{n-2}/a_n
(v) Sum of products of roots taking three at a time = $- a_{n-3}/a_n$
(vi) …
(vii) Product of all roots = $(-1)^n a_o/a_n$

Example 20: Given that the equation $x^3 + ax + b = 0$ has roots α, β and γ are all rational numbers. Show that the roots of equation $\alpha x^2 + \beta x + \gamma = 0$ are also rational.

Example 21: The four roots of the equation $2x^4 + mx^2 + 8 = 0$ are all integers.
Find m and the roots of the equation.

Example 22: Show that not all the roots of the equation $x^3 - x^2 - \frac{x}{2} - \frac{1}{6} = 0$ are real.

Example 23: If α, β and γ are the three roots of the equation $x^3 - 6x^2 + kx + k = 0$, find the values of k such that $(\alpha - 1)^3 + (\beta - 2)^3 + (\gamma - 3)^3 = 0$.
For each of the possible values of k, solve the equation.

Example 24: Prove that the roots of the polynomial
$P(x) = x^n - 5x^{n-1} - 12x^{n-2} + 15x^{n-3} + a_{n-4}x^{n-4} + \ldots + a_n$
cannot all be positive real numbers.

Example 25: Given that a, b, c, d, e and f are all real numbers, and that $f(x) = x^8 - 4x^7 + 7x^6 + ax^5 + bx^4 + cx^3 + dx^2 + ex + f$ can be factorized into 8 linear factors of the form $x - xi$, where $xi > 0$ for I = 1, 2, …, 8. Find all the possible values of f.
(2003 15th APMO No. 1)